Cyclohexane Reactor Ver 1 · operator training simulator

Instrument list, I/O count and instrument sizing

Document no. CHX-V1-IN-INL-001 · **Discipline** Instrumentation and control

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

- 146 tags: 76 in the simulator, 50 added on the EPC P&ID CHX-V1-PID-001, 20 recommended by the safety studies.
- I/O: 168 channels (DCS 110, SIS 58) + 37 spare (20 %) = 205; without the recommended items 132.
- 11 control valves sized (IEC 60534-2-1) for the three HMB cases; 3 orifices (ISO 5167-2); 3 analysers.
- 9 engineering queries (EQ-ei-1 ... EQ-ei-9). Every value not taken from the model is labelled assumed or typical practice (section 9).

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Basis and legend

Item	Basis
Instrument signals	4–20 mA HART for analogue; 24 V DC dry contacts for status; solenoids 24 V DC, de-energise to trip (assumed, typical)
Systems	DCS for control, indication and BPCS on/off valves; SIS-100 (safety PLC) for trips, permissives and SIS valves; DCS ↔ SIS serial link for repeats (typical)
Spare I/O	20 % installed spare per system and signal type, rounded up (typical practice)
Ranges	Existing tags: the model's ranges (makeCtrls() lo/hi, IND r in ui1.js). New tags: the range of the DCS display they feed; recommended tags by the rule normal at 50–70 % of span
Instrument types	Typical choices for the service (fluid, temperature, slurry, SIS diversity); the model does not fix them except the R-101 elements, which the model names thermocouples
Valve sizing	IEC 60534-2-1 (ISA-75.01) liquid and gas equations; Fp = 1 (body = line or reducer losses neglected); FL, xT, xFz typical for the valve style; equal % rangeability 50; rated Cv from typical body Cvs and reduced trims (none for slurry); governing case ≤ 85 % open; every case 10–90 %
Valve ΔP	From CHX-V1-PE-LL-001 part 6 (hydraulic budgets) and the model's pressures at each HMB case; line losses of the line list removed; TV-102 jacket pressure = saturation at T_2 + Q/UAj (model); R-102 jacket CW case 2 bar across the valve (assumed)
Critical pressures	benzene 48.9, cyclohexane 40.7, water 220.6 bar a (typical data); vapour pressures from the model (PsatC, PsatW)
Flow element	ISO 5167-2 orifice, flange tappings, Reader-Harris/Gallagher C; ΔP at the top of the transmitter range 250 mbar (typical), 1 000 mbar for FE-101 to keep β ≤ 0.75; gas expansibility ε (ISO 5167-2)
Steam and gas	k = 1.30 (steam), 1.40 (hydrogen-rich gas); steam Z = 0.95 (eproc); ideal gas density at the valve inlet

1.1 · Columns of the instrument list

Column	Meaning
Tag	ISA-5.1 letters + loop number. DCS functions (FIC, TI, ...) carry no hardware; field instruments (FT, TT, PT, LT, AT, ...) own the input channel; valves own the output.
Type	Category and the typical choice of measuring principle for the service (typical practice; the model fixes the tag, not the device).
P&ID sheet	Sheet of CHX-V1-PID-001 where the bubble or valve is drawn; "(note)" = named in a note only; "–" = not drawn.
Line / equipment	Line number of the line list CHX-V1-PE-LL-001, or the equipment item.
Range	Calibrated range: model ranges (makeCtrls() lo/hi, IND r of ui1.js); new tags inherit the range of the DCS display they feed; recommended tags by the rule.
Normal	Model at design (snapDesign, noise off).
Alarms	ALM (ui1.js) with the alarm ids of the C&E sheet.
Trip	Settings K.T1hh, K.T3hh, K.P1hh, K.L4hh and delays TRIPDLY (model.js); recommended trips with assumed settings.
I/O	Hardware channels: system (DCS / SIS) and type (AI, AO, DI, DO).
Fail	Control valves from the FAIL map of step(); on/off valves from the EPC P&ID register.
Status	model = tag exists in the simulator; EPC P&ID = added on the EPC P&ID; rec. = recommended (Safety layers, HAZOP, LOPA), not in the model.

1.2 · Reference documents

Document	Title
CHX-V1-PID-001	EPC P&ID sheets 1–2 (tab P&ID, alternative version)
CHX-V1-IN-CE-001	Cause and effect matrix (tab C&E matrix)
CHX-V1-IN-SRS-001	Safety requirements specification (tab SRS of SIF)
CHX-V1-PE-HMB-001	Heat and material balance (cases D/H/L)

Document	Title
CHX-V1-PE-LL-001	Line list and line sizing
CHX-V1-EE-LL-001	Electrical load list (UPS loads of DCS/SIS)

2 · Instrument list

One row per tag in loop order. Normal values are the simulator at design; types are typical choices. The same list is in the Instrumentation & control tab with filters and a CSV export.

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
1	**AI-100**	Loop oxygen (start-up purge analyzer; < 1 % before hydrogen)	**DCS function** DCS indicator	2	8"-H-2007-B2	0–21	% O2	0.00	H 1 AAH-100		–	—	model
2	**AT-100**	Loop oxygen (start-up purge analyzer; < 1 % before hydrogen)	**Analyser** Paramagnetic oxygen analyser	2	8"-H-2007-B2	0–21	% O2	0.00			SIS AI	—	EPC P&ID
3	**AT-100B**	Second loop O_2 analyser (1oo2 with AT-100 on the hydrogen permissive)	**Analyser** Paramagnetic oxygen analyser	1 (note)	8"-H-2007-B2	0–21	% O_2	0.0			SIS AI	—	rec.
4	**EIV-100**	Benzene battery limit emergency isolation valve (fire-safe)	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC, fire-safe, remote from the control room	1	6"-P-1001-B1	–		–			SIS DO + 2 DI	FC	rec.
5	**HS-100**	Emergency shutdown pushbutton (unit trip + EDV-101)	**Hand switch** Hard-wired ESD pushbutton	2 (note)	control-room console	–		released			SIS DI	—	model
6	**P-100A**	Benzene feed pump P-100A	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	1	MCC	–		running			DCS 2 DI + 2 DO	—	model
7	**P-100B**	Benzene feed pump P-100B (spare)	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	1	MCC	–		stopped (standby)			DCS 2 DI + 2 DO	—	model
8	**SIS-100**	SIS logic solver: 6 trip inputs + ESD, latched first-out, permissives DCS ↔ SIS by a serial link (status, first-out, repeats of SIS transmitters); no SIL credit for the link.	**Logic** Safety PLC separate from the DCS (own processor, I/O, power), SIL 3 capable (assumed, C&E / SRS)	1 (note)	control room / equipment room	–		healthy			–	—	model
9	**TI-100**	R-101 temperature, independent element (TSHH-100 trips the unit at 220 °C)	**DCS function** DCS indicator	1	R-101	0–300	°C	200.0	H 210 TAH-100		–	—	model
10	**TSHH-100**	TSHH-100 R-101 temperature high-high	**SIS function** SIS logic function on TT-100	1	R-101	0–300	°C	200.0		HH 220 °C · 5 s	–	—	model
11	**TT-100**	R-101 temperature, independent element (TSHH-100 trips the unit at 220 °C) Second, separate element (TE-100) in its own thermowell; SIS sensor of TSHH-100. The model calls the R-101 elements thermocouples.	**Temperature** Thermocouple type K in thermowell + head transmitter	1	R-101	0–300	°C	200.0			SIS AI	—	EPC P&ID

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
12	**EDV-101**	Emergency depressuring valve, R-102 off-gas to flare Opened by the ESD (HS-100) only; fail position on the P&ID "FO (assumed)".	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	8"-FL-3013-A2	–	open / closed	closed			SIS DO + 2 DI	FO	model
13	**EIV-101**	R-101 bottom emergency isolation valve (fire-safe)	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC, fire-safe, remote from the control room	1	18"-P-1003-B3	–		–			SIS DO + 2 DI	FC	rec.
14	**FIC-101**	Benzene feed to R-101 → FV-101 (fail closed)	**DCS function** DCS PID controller → FV-101	1	3"-P-1002-B1	0–60	t/h	34.8	L 25 · H 42 FAL-101, FAH-101		–	—	model
15	**FT-101**	Benzene feed to R-101 → FV-101 (fail closed)	**Flow** DP flow: orifice plate (ISO 5167-2, flange taps) + DP transmitter, square-root extraction	1	3"-P-1002-B1	0–60	t/h	34.8			DCS AI	—	EPC P&ID
16	**FV-101**	Benzene feed to R-101 Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, contoured plug · equal % · Cv 69.3 (3" body)	1	3"-P-1002-B1	0–100	% open	61.9			DCS AO	FC	model
17	**HS-101**	E-101 HEAT / COOL selector (moves HY-101A…D)	**Hand switch** DCS selector (soft)	1	DCS	–		COOL			–	—	model
18	**HY-101A**	D-101 downcomer (HS-101 COOL) Moved by HS-101 (HEAT/COOL); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	1	8"-BFW-3005-A1	–	open / closed	open (COOL)			DCS DO + 2 DI	FO	EPC P&ID
19	**HY-101B**	E-101 riser (HS-101 COOL) Moved by HS-101 (HEAT/COOL); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	1	16"-S-3006-A1	–	open / closed	open (COOL)			DCS DO + 2 DI	FO	EPC P&ID
20	**HY-101C**	E-101 condensate (HS-101 HEAT) Moved by HS-101 (HEAT/COOL); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	1	E-101 shell condensate	–	open / closed	closed			DCS DO + 2 DI	FC	EPC P&ID
21	**HY-101D**	MP steam to E-101 shell (HS-101 HEAT; closed by unit trip) Moved by HS-101 (HEAT/COOL); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	1	8"-S-3002-A1	–	open / closed	closed			DCS DO + 2 DI	FC	EPC P&ID
22	**JI-101**	Agitator M-101 power (design ≈150 kW gassed)	**DCS function** DCS indicator	1	M-101 (MCC)	0–300	kW	153	L 80 JAL-101		–	—	model
23	**JT-101**	Agitator M-101 power (design ≈150 kW gassed) Agitator power from the motor (MCC power transducer).	**Power** Motor power transducer in the MCC	1	M-101 (MCC)	0–300	kW	153			DCS AI	—	EPC P&ID

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
24	**K-101**	Recycle hydrogen compressor K-101	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	2	MCC	–		running			DCS 2 DI + 2 DO	—	model
25	**LIC-101**	R-101 level → LV-101 overflow to R-102 (fail closed)	**DCS function** DCS PID controller → LV-101	1	R-101	0–100	%	65.0	L 45 · H 82 LAL-101, LAH-101		–	—	model
26	**LSLL-101**	R-101 level low-low → unit trip	**SIS function** SIS logic function on LT-101S	1 (note)	R-101	0–100	%	65.0		LL 30 %	–	—	rec.
27	**LT-101**	R-101 level → LV-101 overflow to R-102 (fail closed) The model's level includes the gas hold-up (froth surface), which a radar sees; a DP cell would read the liquid mass only (lower by the hold-up).	**Level** Non-contact radar (80 GHz) level transmitter	1	R-101	0–100	%	65.0			DCS AI	—	EPC P&ID
28	**LT-101S**	R-101 level, separate SIS element of LSLL-101	**Level** Guided-wave radar level transmitter	1 (note)	R-101	0–100	%	65.0			SIS AI	—	rec.
29	**LV-101**	R-101 overflow slurry to R-102 Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · eccentric rotary plug (slurry) · linear · Cv 110 (3" body)	1	4"-P-1008-B3	0–100	% open	50.0			DCS AO	FC	model
30	**M-101**	R-101 agitator motor M-101	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	1	MCC	–		running			DCS 2 DI + 2 DO	—	model
31	**P-101A**	R-101 pump-around pump P-101A	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	1	MCC	–		running			DCS 2 DI + 2 DO	—	model
32	**P-101B**	R-101 pump-around pump P-101B (spare)	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	1	MCC	–		stopped (standby)			DCS 2 DI + 2 DO	—	model
33	**PI-101**	R-101 pressure (PSHH-101 trips at 33.5 bar g)	**DCS function** DCS indicator	1	R-101 overhead	0–40	bar g	30.00			–	—	model
34	**PIAH-101**	R-101 pressure pre-alarm on PT-101 (DCS soft alarm)	**DCS function** DCS alarm on the PI-101 repeat	1 (note)	R-101 overhead	0–40	bar g	30.0	H 32.5 PAH-101		–	—	rec.
35	**PSHH-101**	PSHH-101 R-101 pressure high-high	**SIS function** SIS logic function on PT-101	1	R-101 overhead	0–40	bar g	30.00		HH 33.5 bar g · 3 s	–	—	model
36	**PT-101**	R-101 pressure (PSHH-101 trips at 33.5 bar g) SIS transmitter of PSHH-101; the DCS reads PI-101 over the SIS link.	**Pressure** Gauge-pressure transmitter	1	R-101 overhead	0–40	bar g	30.00			SIS AI	—	EPC P&ID
37	**SSL-101**	Agitator shaft speed (diverse from the M-101 run contact)	**Switch / status** Shaft-speed switch (proximity probe)	1	M-101 / R-101 agitator shaft	–		–			SIS DI	—	rec.

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
38	**TDA-101**	Deviation alarm TIC-101 element vs TI-100 element	**DCS function** DCS calculation + alarm |TIC-101 − TI-100|	1 (note)	R-101	0–20	°C	0.0	H 5 TDAH-101		–	—	rec.
39	**TIC-101**	R-101 temperature → cascade to FIC-104 pump-around through E-101	**DCS function** DCS PID controller (cascade master of FIC-104)	1	R-101	0–300	°C	200.0	L 190 · H 208 TAL-101, TAH-101		–	—	model
40	**TT-101**	R-101 temperature → cascade to FIC-104 pump-around through E-101 Control element TE-101, thermocouple (as named in the model).	**Temperature** Thermocouple type K in thermowell + head transmitter	1	R-101	0–300	°C	200.0			DCS AI	—	EPC P&ID
41	**XS-101**	XS-101 agitator M-101 stopped with XV-101 open	**SIS function** SIS logic function on a motor run contact (status, 1oo1)	1	M-101	–		running		motor stopped · 10 s	SIS DI	—	model
42	**XV-101**	Benzene feed SIS shut-off valve SIS final element of every unit trip; opens only through the start permissives.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	1	3"-P-1002-B1	–	open / closed	open			SIS DO + 2 DI	FC	model
43	**EIV-102**	R-102 bottom emergency isolation valve (fire-safe)	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC, fire-safe, remote from the control room	2	4"-P-1010-B3	–		–			SIS DO + 2 DI	FC	rec.
44	**FI-102**	R-102 liquid to R-103 through F-102	**DCS function** DCS indicator	2	3"-P-1012-B1	0–60	t/h	20.9			–	—	model
45	**FT-102**	R-102 liquid to R-103 through F-102 Mass flow and density at 160 °C; clean liquid after the candle filter.	**Flow** Coriolis mass flowmeter	2	3"-P-1012-B1	0–60	t/h	20.9			DCS AI	—	EPC P&ID
46	**HS-102**	R-102 jacket HEAT / COOL selector (moves HY-102A…D)	**Hand switch** DCS selector (soft)	2	DCS	–		HEAT			–	—	model
47	**HY-102A**	R-102 jacket MP steam (HS-102 HEAT) Moved by HS-102 (HEAT/COOL); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	6"-S-3003-A1	–	open / closed	open (HEAT)			DCS DO + 2 DI	FC	EPC P&ID
48	**HY-102B**	R-102 jacket CW supply (HS-102 COOL) Moved by HS-102 (HEAT/COOL); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	6"-CW-3011-A1	–	open / closed	closed			DCS DO + 2 DI	FO	EPC P&ID
No	**Tag**	**Service**	**Type**	**Sheet**	**Line / equipment**	**Range**	**Unit**	**Normal**	**Alarms**	**Trip**	**I/O**	**Fail**	**Status**
49	**HY-102C**	R-102 jacket CW return (HS-102 COOL) Moved by HS-102 (HEAT/COOL); the	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches	2	R-102 jacket CW return	–	open / closed	closed			DCS DO + 2 DI	FO	EPC P&ID

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
		model switches the routes without these valves.	ZSO/ZSC										
50	**HY-102D**	R-102 jacket condensate (HS-102 HEAT) Moved by HS-102 (HEAT/COOL); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	R-102 jacket condensate	–	open / closed	open (HEAT)			DCS DO + 2 DI	FC	EPC P&ID
51	**LIC-102**	R-102 level → LV-102 to R-103 through filter F-102 (fail closed)	**DCS function** DCS PID controller → LV-102	2	R-102	0–100	%	55.0	L 30 · H 80 LAL-102, LAH-102		–	—	model
52	**LT-102**	R-102 level → LV-102 to R-103 through filter F-102 (fail closed) As LT-101 (gassed slurry).	**Level** Non-contact radar (80 GHz) level transmitter	2	R-102	0–100	%	55.0			DCS AI	—	EPC P&ID
53	**LV-102**	R-102 liquid through F-102 to R-103 Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, contoured plug · linear · Cv 27.5 (3" body)	2	3"-P-1012-B1	0–100	% open	50.0			DCS AO	FC	model
54	**M-102**	R-102 agitator motor (drawn on the P&ID, not a motor in the model)	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	2	MCC	–		–			DCS 2 DI + 2 DO	—	EPC P&ID
55	**P-102A**	R-102 transfer pump P-102A	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	2	MCC	–		running			DCS 2 DI + 2 DO	—	model
56	**P-102B**	R-102 transfer pump P-102B (spare)	**Motor interface** MCC interface: run DI, trip DI, start DO, stop DO	2	MCC	–		stopped (standby)			DCS 2 DI + 2 DO	—	model
57	**PDI-102**	Catalyst filter F-102 differential pressure	**DCS function** DCS indicator	2	F-102A/B	0–4	bar	0.26	H 0.8 PDAH-102		–	—	model
58	**PDT-102**	Catalyst filter F-102 differential pressure Across the running candle filter; remote seals (slurry side).	**Pressure** Differential-pressure transmitter with remote diaphragm seals	2	F-102A/B	0–4	bar	0.26			DCS AI	—	EPC P&ID
59	**PI-102**	R-102 pressure	**DCS function** DCS indicator	2	R-102	0–40	bar g	29.60			–	—	model
60	**PT-102**	R-102 pressure	**Pressure** Gauge-pressure transmitter	2	R-102	0–40	bar g	29.60			DCS AI	—	EPC P&ID
61	**TIC-102**	R-102 temperature → TV-102 jacket (MP steam in HEAT, CW in COOL; fail closed)	**DCS function** DCS PID controller → TV-102	2	R-102	0–300	°C	160.0	L 148 · H 172 TAL-102, TAH-102		–	—	model
62	**TT-102**	R-102 temperature → TV-102 jacket (MP steam in HEAT, CW in COOL; fail closed)	**Temperature** RTD Pt100 (3-wire) in thermowell + head transmitter	2	R-102	0–300	°C	160.0			DCS AI	—	EPC P&ID
No	**Tag**	**Service**	**Type**	**Sheet**	**Line / equipment**	**Range**	**Unit**	**Normal**	**Alarms**	**Trip**	**I/O**	**Fail**	**Status**
63	**TV-102**	MP steam to the R-102 jacket (HEAT)	**Control valve** Control valve · globe, contoured	2	6"-S-3003-A1	0–100	% open	50.0			DCS AO	FC	model

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
		Positioner with HART travel feedback (no extra I/O).	plug · equal % · Cv 69.3 (3" body)										
64	**XS-102**	XS-102 compressor K-101 stopped with XV-101 open	**SIS function** SIS logic function on a motor run contact (status, 1oo1)	2	K-101	–		running		motor stopped · 10 s	SIS DI	—	model
65	**AI-103**	Product benzene, on-line GC (4-min cycle, 2-min sample delay; spec ≤ 10 ppm)	**DCS function** DCS indicator	2	4"-P-1013-B1	0–100	ppm	2.0	H 10 AAH-103		–	—	model
66	**AT-103**	Product benzene, on-line GC (4-min cycle, 2-min sample delay; spec ≤ 10 ppm) One GC, two components: benzene (AI-103) and methylcyclopentane (AI-104).	**Analyser** Process gas chromatograph (FID), product liquid	2	4"-P-1013-B1	0–100	ppm	AI-103 2.0 ppm · AI-104 60 ppm			DCS 2 AI + DI	—	EPC P&ID
67	**FIC-103**	Hydrogen (recycle + make-up) to R-101 sparger → FV-103 (fail closed)	**DCS function** DCS PID controller → FV-103	1	8"-H-2008-B2	0–80 000	Nm³/h	46 000	L 35 000 FAL-103		–	—	model
68	**FSLL-103**	Hydrogen flow low-low → XV-101 shut + feed pumps stop (replaces XS-102)	**SIS function** SIS logic function on FT-103S	1 (note)	8"-H-2008-B2	0–80 000	Nm³/h	46 000.0		LL 15 000 Nm³/h	–	—	rec.
69	**FT-103**	Hydrogen (recycle + make-up) to R-101 sparger → FV-103 (fail closed)	**Flow** DP flow: orifice plate (ISO 5167-2, flange taps) + DP transmitter, square-root extraction	1	8"-H-2008-B2	0–80 000	Nm³/h	46 000			DCS AI	—	EPC P&ID
70	**FT-103S**	Hydrogen to R-101, separate SIS flow element of FSLL-103	**Flow** DP flow: orifice plate (ISO 5167-2, flange taps) + DP transmitter, square-root extraction	1 (note)	8"-H-2008-B2	0–80 000	Nm³/h	46 000.0			SIS AI	—	rec.
71	**FV-103**	Recycle + make-up hydrogen to the R-101 sparger Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, contoured plug · equal % · Cv 400 (6" body)	1	8"-H-2008-B2	0–100	% open	50.0			DCS AO	FC	model
72	**LG-103**	V-103 reflux drum level, local Local gauge, no signal.	**Level** Magnetic level gauge (local)	1	V-103	–		–			–	—	EPC P&ID
73	**TI-103**	E-103 reflux condenser outlet / V-103 (reflux and vent gas temperature)	**DCS function** DCS indicator	1	V-103 / E-103 outlet	0–250	°C	55.7	H 75 TAH-103		–	—	model
74	**TT-103**	E-103 reflux condenser outlet / V-103 (reflux and vent gas temperature)	**Temperature** RTD Pt100 (3-wire) in thermowell + head transmitter	1	V-103 / E-103 outlet	0–250	°C	55.7			DCS AI	—	EPC P&ID
75	**AI-104**	Product methylcyclopentane (isomer made when R-101 runs hot; spec ≤ 200 ppm)	**DCS function** DCS indicator	2	4"-P-1013-B1	0–500	ppm	60	H 200 AAH-104		–	—	model

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
76	**EIV-104**	V-104 product emergency isolation valve (fire-safe)	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC, fire-safe, remote from the control room	2	4"-P-1013-B1	–		–			SIS DO + 2 DI	FC	rec.
77	**FIC-104**	Pump-around flow through E-101 → FV-104 (fail open)	**DCS function** DCS PID controller → FV-104	1	16"-P-1005-B3	0–700	t/h	357			–	—	model
78	**FT-104**	Pump-around flow through E-101 → FV-104 (fail open) Not an orifice: Raney-Ni slurry erodes and plugs a sharp edge; cyclohexane is non-conductive, so no magnetic meter.	**Flow** DP flow: wedge element + DP transmitter with flushed remote seals (catalyst slurry)	1	16"-P-1005-B3	0–700	t/h	357			DCS AI	—	EPC P&ID
79	**FV-104**	Pump-around to E-101 (three-way, E-101 port) Positioner with HART travel feedback (no extra I/O).	**Control valve** Three-way control valve · three-way globe, two ports · linear · Cv 1100 (10" body)	1	16"-P-1005-B3	0–100	% open	45.4			DCS AO	FO	model
80	**LIC-104**	V-104 separator level → LV-104 product to storage (fail closed)	**DCS function** DCS PID controller → LV-104	2	V-104	0–100	%	50.0	L 25 · H 78 LAL-104, LAH-104		–	—	model
81	**LSHH-104**	LSHH-104 V-104 level high-high	**SIS function** SIS logic function on LT-104S	2	V-104	0–100	%	50.0		HH 90 % · 5 s	–	—	model
82	**LSLL-104**	V-104 level low-low → closes XV-104 (liquid seal)	**SIS function** SIS logic function on LT-104S	2	V-104	0–100	%	50.0		LL 10 %	–	—	rec.
83	**LT-104**	V-104 separator level → LV-104 product to storage (fail closed) Control transmitter of LIC-104.	**Level** DP level transmitter (wet leg / remote seals)	2	V-104	0–100	%	50.0			DCS AI	—	EPC P&ID
84	**LT-104S**	V-104 level, SIS element of LSHH-104 Separate SIS element of LSHH-104 (model meas.LI104T), different technology from LT-104.	**Level** Guided-wave radar level transmitter	2	V-104	0–100	%	50.0			SIS AI	—	EPC P&ID
85	**LV-104**	Cyclohexane product to storage Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, multi-stage anti-cavitation trim · linear · Cv 17.6 (1-1/2" body)	2	4"-P-1013-B1	0–100	% open	50.0			DCS AO	FC	model
86	**TI-104**	Pump-around return from E-101 to R-101	**DCS function** DCS indicator	1	16"-P-1006-B3	0–300	°C	159.3			–	—	model
87	**TT-104**	Pump-around return from E-101 to R-101	**Temperature** RTD Pt100 (3-wire) in thermowell + head transmitter	1	16"-P-1006-B3	0–300	°C	159.3			DCS AI	—	EPC P&ID
88	**XV-104**	V-104 product SIS shut-off valve (LSLL-104)	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	4"-P-1013-B1	–		–			SIS DO + 2 DI	FC	rec.

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
89	**AI-105**	Recycle gas hydrogen purity (rest CH4 / N2)	**DCS function** DCS indicator	2	8"-H-2007-B2	50–100	% H2	92.1	L 85 AAL-105		–	—	model
90	**AT-105**	Recycle gas hydrogen purity (rest CH4 / N2)	**Analyser** Thermal-conductivity hydrogen analyser	2	8"-H-2007-B2	50–100	% H2	92.1			DCS AI + DI	—	EPC P&ID
91	**FI-105**	Reflux from V-103 back to R-101 (the product R-101 boils off and condenses)	**DCS function** DCS indicator	1	8"-P-1009-B1	0–150	t/h	46.5			–	—	model
92	**FT-105**	Reflux from V-103 back to R-101 (the product R-101 boils off and condenses) Gravity reflux line: a vortex meter takes little pressure.	**Flow** Vortex flowmeter	1	8"-P-1009-B1	0–150	t/h	46.5			DCS AI	—	EPC P&ID
93	**PIC-105**	Loop pressure at V-104 → make-up hydrogen PV-105 (fail closed)	**DCS function** DCS PID controller → PV-105	2	V-104	0–40	bar g	29.00	L 27.5 · H 30.5 PAL-105, PAH-105		–	—	model
94	**PT-105**	Loop pressure at V-104 → make-up hydrogen PV-105 (fail closed)	**Pressure** Gauge-pressure transmitter	2	V-104	0–40	bar g	29.00			DCS AI	—	EPC P&ID
95	**PV-105**	Make-up hydrogen to the loop Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, contoured plug · equal % · Cv 69.3 (3" body)	2	6"-H-2010-B1	0–100	% open	50.0			DCS AO	FC	model
96	**XV-105**	Make-up hydrogen SIS shut-off valve Opens only with loop O_2 < 1 %.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	6"-H-2010-B1	–	open / closed	open			SIS DO + 2 DI	FC	model
97	**XV-105B**	Second make-up hydrogen SIS valve, in series with XV-105	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	1 (note)	6"-H-2010-B1	–		–			SIS DO + 2 DI	FC	rec.
98	**FI-106**	Make-up hydrogen through PV-105	**DCS function** DCS indicator	2	6"-H-2010-B1	0–60 000	Nm³/h	31 974			–	—	model
99	**FT-106**	Make-up hydrogen through PV-105 Hydrogen consumption = make-up: compensated for P and T.	**Flow** DP flow: orifice plate + DP transmitter, pressure and temperature compensated	2	6"-H-2010-B1	0–60 000	Nm³/h	31 974			DCS AI	—	EPC P&ID
100	**TI-106**	R-103 inlet (R-102 liquid + vapour + hydrogen)	**DCS function** DCS indicator	2	R-103 inlet	0–250	°C	158.5			–	—	model
101	**TT-106**	R-103 inlet (R-102 liquid + vapour + hydrogen)	**Temperature** RTD Pt100 (3-wire) in thermowell + head transmitter	2	R-103 inlet	0–250	°C	158.5			DCS AI	—	EPC P&ID
102	**FI-107**	Purge gas to fuel gas through HIC-107	**DCS function** DCS indicator	2	1-1/2"-H-2011-B2	0–6 000	Nm³/h	2 017			–	—	model
103	**FT-107**	Purge from V-104 to fuel gas (hand controller, fail closed)	**Flow** Vortex flowmeter	2	1-1/2"-H-2011-B2	0–6 000	Nm³/h	2 017			DCS AI	—	EPC P&ID

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
104	**HIC-107**	Purge from V-104 to fuel gas (hand controller, fail closed)	**DCS function** DCS hand controller (HIC) → HV-107	2	1-1/2"-H-2011-B2	0–100	%	output 25 % (purge 2017 Nm³/h)			–	—	model
105	**HV-107**	Purge from V-104 to fuel gas Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, cage-guided low-noise trim · linear · Cv 3 (1" body)	2	1-1/2"-H-2011-B2	0–100	% open	25.0			DCS AO	FC	model
106	**TI-107**	R-103 hottest bed thermocouple (TSHH-107 at 220 °C)	**DCS function** DCS calculation: highest of TI-107A…E	2 (note)	R-103 cells 1–5	100–250	°C	158.6	H 185 TAH-107		–	—	model
107	**TI-107A**	R-103 bed thermocouple, top (cell 1)	**DCS function** DCS indicator	2	R-103 cell 1 (top)	100–250	°C	158.5			–	—	model
108	**TI-107B**	R-103 bed thermocouple, cell 2	**DCS function** DCS indicator	2	R-103 cell 2	100–250	°C	158.6			–	—	model
109	**TI-107C**	R-103 bed thermocouple, middle (cell 3)	**DCS function** DCS indicator	2	R-103 cell 3	100–250	°C	158.6			–	—	model
110	**TI-107D**	R-103 bed thermocouple, cell 4	**DCS function** DCS indicator	2	R-103 cell 4	100–250	°C	158.6			–	—	model
111	**TI-107E**	R-103 bed thermocouple, bottom (cell 5 = outlet)	**DCS function** DCS indicator	2	R-103 cell 5 (bottom)	100–250	°C	158.6			–	—	model
112	**TSHH-107**	TSHH-107 R-103 bed temperature high-high	**SIS function** SIS logic function on TT-107A…E	2	R-103 cells 1–5	100–250	°C	158.6		HH 220 °C · 5 s	–	—	model
113	**TT-107A**	R-103 bed thermocouple, top (cell 1) One multipoint assembly TE-107A…E (5 cells); also the SIS sensors of TSHH-107.	**Temperature** Thermocouple type K, one point of a multipoint bed assembly + transmitter	2	R-103 cell 1 (top)	100–250	°C	158.5			SIS AI	—	EPC P&ID
114	**TT-107B**	R-103 bed thermocouple, cell 2	**Temperature** Thermocouple type K, one point of a multipoint bed assembly + transmitter	2	R-103 cell 2	100–250	°C	158.6			SIS AI	—	EPC P&ID
115	**TT-107C**	R-103 bed thermocouple, middle (cell 3)	**Temperature** Thermocouple type K, one point of a multipoint bed assembly + transmitter	2	R-103 cell 3	100–250	°C	158.6			SIS AI	—	EPC P&ID
116	**TT-107D**	R-103 bed thermocouple, cell 4	**Temperature** Thermocouple type K, one point of a multipoint bed assembly + transmitter	2	R-103 cell 4	100–250	°C	158.6			SIS AI	—	EPC P&ID
117	**TT-107E**	R-103 bed thermocouple, bottom (cell 5 = outlet)	**Temperature** Thermocouple type K, one point of a multipoint bed assembly + transmitter	2	R-103 cell 5 (bottom)	100–250	°C	158.6			SIS AI	—	EPC P&ID

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
118	**TT-107S** ×10	R-103 separate SIS elements, two per cell (2oo3 per cell with TT-107A…E)	**Temperature** Thermocouple type K, one point of a multipoint bed assembly + transmitter	—	R-103 (2 per cell)	100–250	°C	158.6			SIS 10 AI	—	rec.
119	**FI-108**	MP steam to the R-102 jacket (HEAT mode)	**DCS function** DCS indicator	2	6"-S-3003-A1	0–6	t/h	2.33			–	—	model
120	**FT-108**	MP steam to the R-102 jacket (HEAT mode)	**Flow** Vortex flowmeter, steam, pressure compensated	2	6"-S-3003-A1	0–6	t/h	2.33			DCS AI	—	EPC P&ID
121	**FI-109**	Boiler feed water to D-101	**DCS function** DCS indicator	1	3"-BFW-3004-A1	0–40	t/h	14.9			–	—	model
122	**FT-109**	Boiler feed water to D-101	**Flow** DP flow: orifice plate (ISO 5167-2, flange taps) + DP transmitter, square-root extraction	1	3"-BFW-3004-A1	0–40	t/h	14.9			DCS AI	—	EPC P&ID
123	**LIC-109**	D-101 steam drum level → LV-109 boiler feed water (fail open)	**DCS function** DCS PID controller → LV-109	1	D-101	0–100	%	50.0	L 25 · H 75 LAL-109, LAH-109		–	—	model
124	**LSLL-109**	D-101 level low-low → unit trip (E-101 water side)	**SIS function** SIS logic function on LT-109S	2 (note)	D-101	0–100	%	50.0		LL 10 %	–	—	rec.
125	**LT-109**	D-101 steam drum level → LV-109 boiler feed water (fail open) Drum level: density compensation at 4.5 bar g (typical three-element drum practice).	**Level** DP level transmitter, density-compensated (drum)	1	D-101	0–100	%	50.0			DCS AI	—	EPC P&ID
126	**LT-109S**	D-101 level, separate SIS element of LSLL-109	**Level** Guided-wave radar level transmitter	—	D-101	0–100	%	50.0			SIS AI	—	rec.
127	**LV-109**	Boiler feed water to D-101 Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, contoured plug · linear · Cv 11.2 (1-1/2" body)	1	3"-BFW-3004-A1	0–100	% open	50.0			DCS AO	FO	model
128	**TI-109**	V-104 separator temperature (E-106 outlet)	**DCS function** DCS indicator	2	V-104	0–250	°C	44.9			–	—	model
129	**TT-109**	V-104 separator temperature (E-106 outlet)	**Temperature** RTD Pt100 (3-wire) in thermowell + head transmitter	2	V-104	0–250	°C	44.9			DCS AI	—	EPC P&ID
130	**FI-110**	Cyclohexane product to storage (HS-115 selects TK-106 on-spec or TK-107 slop)	**DCS function** DCS indicator	2	4"-P-1014-A2	0–60	t/h	37.5			–	—	model
131	**FT-110**	Cyclohexane product to storage (HS-115 selects TK-106 on-spec or TK-107 slop) Mass flow of the product to storage (accounting-grade).	**Flow** Coriolis mass flowmeter	2	4"-P-1014-A2	0–60	t/h	37.5			DCS AI	—	EPC P&ID

No	Tag	Service	Type	Sheet	Line / equipment	Range	Unit	Normal	Alarms	Trip	I/O	Fail	Status
132	**PIC-110**	D-101 steam pressure → PV-110 LP steam export (fail open)	**DCS function** DCS PID controller → PV-110	1	D-101	0–10	bar g	4.50	L 3.8 · H 5.6 PAL-110, PAH-110		–	—	model
133	**PT-110**	D-101 steam pressure → PV-110 LP steam export (fail open)	**Pressure** Gauge-pressure transmitter	1	D-101	0–10	bar g	4.50			DCS AI	—	EPC P&ID
134	**PV-110**	LP steam export from D-101 Positioner with HART travel feedback (no extra I/O).	**Control valve** Control valve · globe, cage-guided low-noise trim · equal % · Cv 1100 (10" body)	1	10"-S-3001-A1	0–100	% open	50.0			DCS AO	FO	model
135	**FI-111**	LP steam exported from D-101 to the 3.5 bar g header	**DCS function** DCS indicator	1	10"-S-3001-A1	0–40	t/h	14.7			–	—	model
136	**FT-111**	LP steam exported from D-101 to the 3.5 bar g header	**Flow** Vortex flowmeter, steam, pressure compensated	1	10"-S-3001-A1	0–40	t/h	14.7			DCS AI	—	EPC P&ID
137	**PI-112**	K-101 discharge (hydrogen header)	**DCS function** DCS indicator	2	8"-H-2008-B2	0–40	bar g	32.00			–	—	model
138	**PT-112**	K-101 discharge (hydrogen header)	**Pressure** Gauge-pressure transmitter	2	8"-H-2008-B2	0–40	bar g	32.00			DCS AI	—	EPC P&ID
139	**HV-113**	E-103 cooling-water supply hand valve (benzene start permissive: open) A hand valve gives the SIS no signal: see ZSO-113.	**Manual valve** Manual gate valve; its position is a start permissive	1	10"-CW-3007-A1	0–100	% open	100			–	—	model
140	**ZSO-113**	HV-113 open limit switch (gives the start permissive a signal)	**Switch / status** Limit switch (valve open)	—	10"-CW-3007-A1	–		–			SIS DI	—	rec.
141	**HS-115**	Product to TK-106 (on spec) / TK-107 (slop), moves HY-115A/B	**Hand switch** DCS selector (soft)	2	DCS	–		TK-106			–	—	model
142	**HY-115A**	product to TK-106 (HS-115) Moved by HS-115 (product tank); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	4"-P-1014-A2	–	open / closed	open (TK-106)			DCS DO + 2 DI	FC	EPC P&ID
143	**HY-115B**	product to TK-107 slop (HS-115) Moved by HS-115 (product tank); the model switches the routes without these valves.	**On/off valve** On/off valve, spring-return actuator, solenoid, limit switches ZSO/ZSC	2	4"-P-1014-A2	–	open / closed	closed			DCS DO + 2 DI	FO	EPC P&ID
144	**TI-120**	Cooling water supply temperature	**DCS function** RTD on the CW supply header (utilities, outside the battery limit) → DCS indicator	2 (note)	CW supply header (outside the battery limit)	20–50	°C	32.0			DCS AI	—	model
145	**I-K101**	SIS stop of K-101 (MCC stop relay)	**SIS function** SIS digital output to the K-101 MCC (de-energise to trip)	2	K-101 (MCC)	–		–			SIS DO	—	model
146	**I-P100**	SIS stop of P-100A/B on unit trip	**SIS function** SIS digital outputs to the P-100A/B MCC	—	P-100A/B (MCC)	–		–			SIS 2 DO	—	rec.

3 · I/O count

Hardware channels only. Field transmitter = 1 AI; control valve = 1 AO (HART positioner feedback, no extra channel); on/off valve = 1 DO (solenoid) + 2 DI (limit switches); motor = 2 DI (running, tripped) + 2 DO (start, stop) on the DCS; SIS sensors, run contacts, trip outputs and SIS valves on SIS-100. Spare 20 % per system and type, rounded up (typical).

3.1 · Including the recommended items

System	AI	AO	DI	DO	Σ used	Spare AI	AO	DI	DO	Σ spare	Inst. AI	AO	DI	DO	Σ installed
DCS	31	11	40	28	110	7	3	8	6	24	38	14	48	34	134
SIS	23	0	23	12	58	5	0	5	3	13	28	0	28	15	71
Total	54	11	63	40	168	12	3	13	9	37	66	14	76	49	205

3.2 · Without the recommended items

System	AI	AO	DI	DO	Σ used	Spare AI	AO	DI	DO	Σ spare	Inst. AI	AO	DI	DO	Σ installed
DCS	31	11	40	28	110	7	3	8	6	24	38	14	48	34	134
SIS	9	0	9	4	22	2	0	2	1	5	11	0	11	5	27
Total	40	11	49	32	132	9	3	10	7	29	49	14	59	39	161

3.3 · By device group

System	Device group	Devices	AI	AO	DI	DO
DCS	Analyser	2	3		2	
DCS	Control valve	11		11		
DCS	DCS function	1	1			
DCS	Flow	11	11			
DCS	Level	4	4			
DCS	Motor interface	9			18	18
DCS	On/off valve	10			20	10
DCS	Power	1	1			
DCS	Pressure	5	5			
DCS	Temperature	6	6			
SIS	Analyser	2	2			
SIS	Flow	1	1			
SIS	Hand switch	1			1	
SIS	Level	3	3			
SIS	On/off valve	9			18	9
SIS	Pressure	1	1			

System	Device group	Devices	AI	AO	DI	DO
SIS	SIS function	4			2	3
SIS	Switch / status	2			2	
SIS	Temperature	16	16			

4 · Control-valve sizing

IEC 60534-2-1 (ISA-75.01). Liquid: Q = N1·C·√(ΔP/(ρ/ρ0)), N1 = 0.865; choked at ΔPmax = FL²(p1 − FF·pv), FF = 0.96 − 0.28√(pv/pc); cavitation index xF = ΔP/(p1 − pv) against xFz. Gas/steam: W = N6·C·Y·√(x·p1·ρ1), N6 = 27.3, x = ΔP/p1, Fγ = k/1.4, Y = 1 − x/(3FγxT), choked at x = FγxT. Opening: linear C/Cr, equal % 1 + ln(C/Cr)/ln 50. Rated Cv = smallest typical body/trim with the governing case ≤ 85 % open (no reduced trim for slurry); every case 10–90 %. Cases from CHX-V1-PE-HMB-001: D design, H 120 %, L 70 %.

4.1 · Summary

Valve	Service	Line	Style · characteristic	Cv D / H / L	Rated Cv · body	Open D / H / L %	10-90	Choked / cavitation	Model Cv 100 %	Model open D	Fail
FV-101	Benzene feed to R-101	3"-P-1002-B1	globe, contoured plug · equal %	20.4 / 26.4 / 13.2	69.3 · 3"	69 / 75 / 58	✓	–	33.0	62	FC
FV-103	Recycle + make-up hydrogen to the R-101 sparger	8"-H-2008-B2	globe, contoured plug · equal %	93.0 / 192 / 47.1	400 · 6"	63 / 81 / 45	✓	–	177	50	FC
PV-105	Make-up hydrogen to the loop	6"-H-2010-B1	globe, contoured plug · equal %	28.6 / 34.0 / 20.5	69.3 · 3"	77 / 82 / 69	✓	–	57.0	50	FC
FV-104	Pump-around to E-101 (three-way, E-101 port)	16"-P-1005-B3	three-way globe, two ports · linear	648 / 822 / 408	1 100 · 10"	59 / 75 / 37	✓	–	1 429	45	FO
LV-101	R-101 overflow slurry to R-102	4"-P-1008-B3	eccentric rotary plug (slurry) · linear	67.8 / 72.9 / 55.9	110 · 3"	62 / 66 / 51	✓	–	129	50	FC
TV-102	MP steam to the R-102 jacket (HEAT)	6"-S-3003-A1	globe, contoured plug · equal %	27.0 / 21.7 / 27.9	69.3 · 3"	76 / 70 / 77	✓	–; COOL (CW) 114 % (EQ-ei-1)	54.0	50	FC
LV-102	R-102 liquid through F-102 to R-103	3"-P-1012-B1	globe, contoured plug · linear	16.9 / 20.5 / 11.6	27.5 · 3"	61 / 75 / 42	✓	–	32.8	50	FC
LV-104	Cyclohexane product to storage	4"-P-1013-B1	globe, multi-stage anti-cavitation trim · linear	9.79 / 11.8 / 6.81	17.6 · 1-1/2"	56 / 67 / 39	✓	cavitation	19.4	50	FC
LV-109	Boiler feed water to D-101	3"-BFW-3004-A1	globe, contoured plug · linear	7.55 / 9.27 / 4.91	11.2 · 1-1/2"	67 / 83 / 44	✓	cavitation	15.0	50	FO
PV-110	LP steam export from D-101	10"-S-3001-A1	globe, cage-guided low-noise trim · equal %	353 / 438 / 227	1 100 · 10"	71 / 76 / 60	✓	–	689	50	FO
HV-107	Purge from V-104 to fuel gas	1-1/2"-H-2011-B2	globe, cage-guided low-noise trim · linear	1.77 / 1.83 / 1.68	3.00 · 1"	59 / 61 / 56	✓	choked	7.08	25	FC

4.2 · Valve by valve

FV-101 · Benzene feed to R-101 (FIC-101)

Line 3"-P-1002-B1 · benzene · globe, contoured plug (FL 0.9, xT 0.7) · equal %: Flow loop on a pump without a defined curve: valve ΔP falls as the flow rises (part of the system loss) → equal %. Rated Cv 69.3 (3" body, 63 % trim), fail FC. Model: FB = K.FBmax · x (0.2 kmol/s at 100 %), independent of ΔP → equivalent Cv at 100 % ≈ 33.0.

Quantity	Design	120 %	70 %
Mass flow kg/h	34 800	41 760	24 360
ρ (inlet) kg/m³	857.8	857.8	857.8
T °C	40.0	40.0	40.0
p1 / p2 bar a	36.00 / 31.47	36.00 / 32.11	36.00 / 30.73
ΔP bar	4.53	3.89	5.27
pv bar a · ΔPmax bar	0.267 · 28.96	0.267 · 28.96	0.267 · 28.96
Cv required	20.4	26.4	13.2
Opening of the rated valve	69 % ✓	75 % ✓	58 % ✓
Model opening	62 %	74 %	43 %

FV-103 · Recycle + make-up hydrogen to the R-101 sparger (FIC-103)

Line 8"-H-2008-B2 · hydrogen-rich gas · globe, contoured plug (FL 0.9, xT 0.7) · equal %: Valve ΔP falls from 3.15 to 0.65 bar between the 70 % and 120 % cases (compressor curve + loop losses ∝ Q²) → equal %. Rated Cv 400 (6" body), fail FC. Model: Q = K.Cv103 · x · √(ΔP) with ΔP = K-101 rise − loop losses (model lumps the sparger) → equivalent Cv at 100 % ≈ 177.

Quantity	Design	120 %	70 %
Mass flow kg/h	6 419	8 216	4 044
ρ (inlet) kg/m³	3.727	3.889	3.439
T °C	60.0	60.0	60.0
p1 / p2 bar a	33.00 / 31.20	32.29 / 31.64	33.83 / 30.68
ΔP bar	1.81	0.65	3.15
x · Y	0.055 · 0.974	0.020 · 0.990	0.093 · 0.956
Cv required	93.0	192	47.1
Opening of the rated valve	63 % ✓	81 % ✓	45 % ✓
Model opening	50 %	92 %	27 %

PV-105 · Make-up hydrogen to the loop (PIC-105)

Line 6"-H-2010-B1 · hydrogen 99.5 % · globe, contoured plug (FL 0.9, xT 0.7) · equal %: Pressure loop on a gas supply: process gain varies with load → equal %. Rated Cv 69.3 (3" body, 63 % trim), fail FC. Model: Qmu = min(K.Cv105 · x · √(Pmu − Ps), K.muMax) → equivalent Cv at 100 % ≈ 57.0.

Quantity	Design	120 %	70 %
Mass flow kg/h	2 976	3 534	2 139
ρ (inlet) kg/m³	2.884	2.884	2.884
T °C	40.0	40.0	40.0
p1 / p2 bar a	36.00 / 30.07	36.00 / 30.09	36.00 / 30.04
ΔP bar	5.93	5.91	5.96
x · Y	0.165 · 0.922	0.164 · 0.922	0.166 · 0.921
Cv required	28.6	34.0	20.5
Opening of the rated valve	77 % ✓	82 % ✓	69 % ✓
Model opening	50 %	59 %	36 %

FV-104 · Pump-around to E-101 (three-way, E-101 port) (FIC-104)

Line 16"-P-1005-B3 · R-101 slurry · three-way globe, two ports (FL 0.85, xT 0.7) · linear: Three-way diverting valve on a constant pump flow: the two ports share the total → linear. Rated Cv 1 100 (10" body), fail FO. Model: Fe = min(Fpa, 1.3 · Fpa · x): E-101 port at full pump flow from 77 %, ΔP not modelled → equivalent Cv at 100 % ≈ 1 429.

Quantity	Design	120 %	70 %
Mass flow kg/h	356 789	452 848	224 215
ρ (inlet) kg/m³	607.3	607.8	606.7
T °C	200.0	200.0	200.0
p1 / p2 bar a	33.00 / 32.33	33.44 / 32.77	32.49 / 31.82
ΔP bar	0.67	0.67	0.67
pv bar a · ΔPmax bar	13.4 · 16.11	13.4 · 16.42	13.4 · 15.74
Cv required	648	822	408
Opening of the rated valve	59 % ✓	75 % ✓	37 % ✓
Model opening	45 %	58 %	28 %

LV-101 · R-101 overflow slurry to R-102 (LIC-101)

Line 4"-P-1008-B3 · R-101 slurry · eccentric rotary plug (slurry) (FL 0.85, xT 0.61) · linear: Level loop with a nearly constant ΔP (R-101 → R-102 + liquid head) → linear. Rated Cv 110 (3" body), fail FC. Model: L1 = K.Cv101 · x · √(P1 − P2 + head) → equivalent Cv at 100 % ≈ 129.

Quantity	Design	120 %	70 %
Mass flow kg/h	36 493	43 644	25 658
ρ (inlet) kg/m³	607.3	607.8	606.7
T °C	200.0	200.0	200.0
p1 / p2 bar a	31.31 / 30.68	31.75 / 30.97	30.80 / 30.34
ΔP bar	0.64	0.79	0.46
pv bar a · ΔPmax bar	13.4 · 14.89	13.4 · 15.21	13.4 · 14.52
Cv required	67.8	72.9	55.9
Opening of the rated valve	62 % ✓	66 % ✓	51 % ✓
Model opening	50 %	53 %	42 %

TV-102 · MP steam to the R-102 jacket (HEAT) (TIC-102)

Line 6"-S-3003-A1 · MP steam 10 bar g · globe, contoured plug (FL 0.9, xT 0.7) · equal %: Temperature loop on steam: the jacket pressure (and so ΔP) moves with the duty → equal %. Rated Cv 69.3 (3" body, 63 % trim), fail FC. Model: steam ≤ K.msj · x (1.293 kg/s at 100 %), duty also limited by UAj·(Tms − T2); ΔP not modelled → equivalent Cv at 100 % ≈ 54.0.

Quantity	Design	120 %	70 %	COOL (CW)
Mass flow kg/h	2 328	1 995	2 378	144 000
ρ (inlet) kg/m³	5.494	5.494	5.494	994.0
T °C	184.0	184.0	184.0	32.0
p1 / p2 bar a	11.01 / 8.74	11.01 / 8.32	11.01 / 8.81	5.01 / 3.01
ΔP bar	2.27	2.69	2.21	2.00
pv bar a · ΔPmax bar	–	–	–	0.0519 · 4.02
x · Y	0.206 · 0.894	0.244 · 0.875	0.200 · 0.897	–
Cv required	27.0	21.7	27.9	118
Opening of the rated valve	76 % ✓	70 % ✓	77 % ✓	114 % ✗
Model opening	50 %	43 %	51 %	100 %

LV-102 · R-102 liquid through F-102 to R-103 (LIC-102)

Line 3"-P-1012-B1 · filtered R-102 liquid · globe, contoured plug (FL 0.9, xT 0.7) · linear: Level loop; valve takes most of the pump differential, ΔP nearly constant → linear. Rated Cv 27.5 (3" body, 25 % trim), fail FC. Model: L2 = √(ΔP/(1/cv² + Rf·clog)), cv = K.Cv102 · x (in series with F-102) → equivalent Cv at 100 % ≈ 32.8.

Quantity	Design	120 %	70 %
Mass flow kg/h	20 937	25 337	14 507
ρ (inlet) kg/m³	641.3	641.3	641.3
T °C	160.0	160.0	160.0
p1 / p2 bar a	33.59 / 30.37	33.70 / 30.53	33.45 / 30.19
ΔP bar	3.21	3.17	3.26
pv bar a · ΔPmax bar	6.60 · 22.68	6.60 · 22.77	6.60 · 22.56
Cv required	16.9	20.5	11.6
Opening of the rated valve	61 % ✓	75 % ✓	42 % ✓
Model opening	50 %	60 %	35 %

LV-104 · Cyclohexane product to storage (LIC-104)

Line 4"-P-1013-B1 · cyclohexane product · globe, multi-stage anti-cavitation trim (FL 0.95, xT 0.8) · linear: Level loop with a large, constant ΔP (29 bar g → tanks) → linear. Rated Cv 17.6 (1-1/2" body, 63 % trim), fail FC. Model: L4 = K.Cv104 · x · √(Ps − Pst) → equivalent Cv at 100 % ≈ 19.4.

Quantity	Design	120 %	70 %
Mass flow kg/h	37 497	44 995	26 244
ρ (inlet) kg/m³	743.0	743.0	743.0
T °C	44.9	44.9	44.9
p1 / p2 bar a	29.97 / 3.59	29.96 / 3.84	29.99 / 3.30
ΔP bar	26.38	26.12	26.69
pv bar a · ΔPmax bar	0.319 · 26.78	0.320 · 26.77	0.319 · 26.80
Cv required	9.79 cav.	11.8 cav.	6.81 cav.
Opening of the rated valve	56 % ✓	67 % ✓	39 % ✓
Model opening	50 %	60 %	35 %

LV-109 · Boiler feed water to D-101 (LIC-109)

Line 3"-BFW-3004-A1 · BFW 105 °C · globe, contoured plug (FL 0.9, xT 0.7) · linear: Drum level on a BFW header of steady pressure → linear. Rated Cv 11.2 (1-1/2" body, 40 % trim), fail FO. Model: mb = K.Cv109 · x · √(Pbfw − Pd) → equivalent Cv at 100 % ≈ 15.0.

Quantity	Design	120 %	70 %
Mass flow kg/h	14 869	18 200	9 700
ρ (inlet) kg/m³	953.5	953.5	953.5
T °C	105.0	105.0	105.0
p1 / p2 bar a	11.00 / 5.57	11.00 / 5.60	11.00 / 5.54
ΔP bar	5.43	5.40	5.46
pv bar a · ΔPmax bar	1.21 · 7.99	1.21 · 7.99	1.21 · 7.99
Cv required	7.55 cav.	9.27 cav.	4.91 cav.
Opening of the rated valve	67 % ✓	83 % ✓	44 % ✓
Model opening	50 %	61 %	33 %

PV-110 · LP steam export from D-101 (PIC-110)

Line 10"-S-3001-A1 · LP steam 4.5 bar g · globe, cage-guided low-noise trim (FL 0.9, xT 0.72) · equal %: Pressure loop, small ΔP (drum − header) that varies with load → equal %. Rated Cv 1 100 (10" body), fail FO. Model: ms = K.Cv110 · x · √(Pd − Phdr) → equivalent Cv at 100 % ≈ 689.

Quantity	Design	120 %	70 %
Mass flow kg/h	14 723	18 020	9 604
ρ (inlet) kg/m³	2.931	2.931	2.931
T °C	155.8	155.8	155.8
p1 / p2 bar a	5.51 / 4.56	5.51 / 4.59	5.51 / 4.53
ΔP bar	0.95	0.92	0.99
x · Y	0.173 · 0.914	0.167 · 0.917	0.179 · 0.911
Cv required	353	438	227
Opening of the rated valve	71 % ✓	76 % ✓	60 % ✓
Model opening	50 %	61 %	33 %

HV-107 · Purge from V-104 to fuel gas (HIC-107)

Line 1-1/2"-H-2011-B2 · loop gas · globe, cage-guided low-noise trim (FL 0.9, xT 0.72) · linear: Hand-set purge with a constant, choked ΔP → linear. Rated Cv 3.00 (1" body, 25 % trim), fail FC. Model: Qpu = K.Cv107 · x · √(Ps − Pfuel) (no choked-flow limit in the model) → equivalent Cv at 100 % ≈ 7.08.

Quantity	Design	120 %	70 %
Mass flow kg/h	281	300	253
ρ (inlet) kg/m³	3.541	3.776	3.188
T °C	44.9	44.9	44.9
p1 / p2 bar a	29.94 / 6.01	29.93 / 6.02	29.95 / 6.01
ΔP bar	23.92	23.92	23.93
x · Y	0.799 · 0.667	0.799 · 0.667	0.799 · 0.667
Cv required	1.77 choked	1.83 choked	1.68 choked
Opening of the rated valve	59 % ✓	61 % ✓	56 % ✓
Model opening	25 %	25 %	25 %

5 · Flow elements (ISO 5167-2 orifice, flange tappings)

qm = C/√(1 − β⁴)·ε·(π/4)·d²·√(2ΔPρ1); C by Reader-Harris/Gallagher; ε = 1 for liquids, 1 − (0.351 + 0.256β⁴ + 0.93β⁸)(1 − (p2/p1)^(1/κ)) for gas. Full scale = the top of the transmitter range.

Element	Service	Line (ID mm)	Full scale	ΔP full scale	β	Bore d mm	C	ε	ReD	Perm. loss	ΔP D / H / L mbar	Checks
FE-101	Benzene feed to R-101	3"-P-1002-B1 (77.92)	60 t/h	1 000 mbar	0.6340	49.40	0.6079	1.0000	546 961	590 mbar	335 / 483 / 164	β 0.1–0.75 ✓ Re ✓ d ≥ 12.5 ✓
FE-103	Recycle + make-up hydrogen to R-101	8"-H-2008-B2 (202.74)	80 000 Nm³/h	250 mbar	0.5883	119.28	0.6044	0.9979	1 933 390	161 mbar	82 / 135 / 33	β 0.1–0.75 ✓ Re ✓ d ≥ 12.5 ✓
FE-109	Boiler feed water to D-101	3"-BFW-3004-A1 (77.92)	40 t/h	250 mbar	0.6967	54.28	0.6079	1.0000	685 111	129 mbar	34 / 51 / 14	β 0.1–0.75 ✓ Re ✓ d ≥ 12.5 ✓

FE-101: 3" line at about 4 m/s at full scale: 250 mbar would need β ≈ 0.81 (> 0.75, outside ISO 5167-2); 1 000 mbar gives β ≈ 0.63 and the 4.5 bar FV-101 margin pays the permanent loss.
FE-103: The whole recycle loop has only ≈ 3 bar of compressor rise: a low range (250 mbar) keeps the permanent loss small.
FE-109: Clean water, ample pressure; 250 mbar is the usual range.

6 · Analyser list

Analyser	Measures	Type (typical choice)	Sample point	Range	Alarm / use	In the model	Cycle · delay	I/O
AT-103	AI-103 benzene · AI-104 methylcyclopentane	Process gas chromatograph, FID, 2 components (typical choice) ppm benzene and isomer need separation: only a GC measures both; it is slow (cycle + transport) and cannot be the measurement of a fast control loop.	4"-P-1013-B1 (V-104 liquid upstream of LV-104), fast loop back to V-104 (typical)	AI-103 0–100 ppm · AI-104 0–500 ppm	AI-103 H 10 ppm (spec) · AI-104 H 200 ppm (spec) Quality monitor and HS-115 tank decision; the fast warning of benzene slip is the R-103 bed temperature rise (≈9 °C per 1 % benzene reacting adiabatically in the bed, model).	Sampled every 60 s into a history; the display is updated every 240 s with the sample taken 120 s before (model step()); V-104 mixing lag τ = n_4/L_4 = 1070 s	4 min · 2 min	2 AI + 1 DI (analyser status)
AT-105	AI-105 recycle hydrogen purity	Thermal-conductivity analyser (H_2 in CH_4/N_2), continuous (typical choice) Hydrogen conducts heat seven times better than methane: a TCD cell reads binary H_2/CH_4 purity directly and fast (T90 ≈ 30 s typical + sample line).	8"-H-2007-B2 (V-104 gas to K-101 suction)	50–100 % H_2	L 85 % H_2; feed permissive ≥ 80 % Purge setting (HIC-107) and the benzene start permissive.	Instantaneous: loop inventory composition (yH2L) with noise; no analyser lag in the model	continuous	1 AI + 1 DI (status)
AT-100	AI-100 loop oxygen	Paramagnetic O_2 analyser 0–21 % (typical choice) Paramagnetic cells are not poisoned by hydrogen and read percent O_2 during the nitrogen purge (a zirconia cell would burn the hydrogen).	8"-H-2007-B2 (loop gas at K-101 suction)	0–21 % O_2	H 1 % with hydrogen admitted; hydrogen permissive < 1 % Start-up purge end point and the hydrogen permissive (SIS); second analyser AT-100B recommended (1oo2).	Instantaneous: nO_2/nG of the loop inventory with noise; no analyser lag in the model	continuous	1 AI (SIS, repeated to the DCS)

AT-103: the screen value is 2 to 6 min old (4 min on average) and V-104 mixes the product with τ ≈ 18 min: a quality monitor and the tank decision (HS-115), not a controlled variable.

7 · Safety instrumented functions (instrument view)

What each trip does is in the cause-and-effect matrix CHX-V1-IN-CE-001 and the requirements of each SIF in CHX-V1-IN-SRS-001; they are not repeated here. This table gives the sensor, the channel and the setting.

Function	Service	Sensor	Setting · delay	Sensor range	Status
TSHH-100	TSHH-100 R-101 temperature high-high	TT-100	HH 220 °C · 5 s	0–300 °C	model
LSLL-101	R-101 level low-low → unit trip	LT-101S	LL 30 %	0–100 %	rec.
PSHH-101	PSHH-101 R-101 pressure high-high	PT-101	HH 33.5 bar g · 3 s	0–40 bar g	model
XS-101	XS-101 agitator M-101 stopped with XV-101 open	M-101 run contact	motor stopped · 10 s	–	model
XS-102	XS-102 compressor K-101 stopped with XV-101 open	K-101 run contact	motor stopped · 10 s	–	model
FSLL-103	Hydrogen flow low-low → XV-101 shut + feed pumps stop (replaces XS-102)	FT-103S	LL 15 000 Nm³/h	0–80 000 Nm³/h	rec.
LSHH-104	LSHH-104 V-104 level high-high	LT-104S	HH 90 % · 5 s	0–100 %	model
LSLL-104	V-104 level low-low → closes XV-104 (liquid seal)	LT-104S	LL 10 %	0–100 %	rec.
TSHH-107	TSHH-107 R-103 bed temperature high-high	TT-107A…E	HH 220 °C · 5 s	100–250 °C	model
LSLL-109	D-101 level low-low → unit trip (E-101 water side)	LT-109S	LL 10 %	0–100 %	rec.
I-K101	SIS stop of K-101 (MCC stop relay)	K-101 (MCC)	–	–	model
I-P100	SIS stop of P-100A/B on unit trip	P-100A/B (MCC)	–	–	rec.

8 · Checks against the model and the other documents

Computed from this document's data source; the tab recomputes them in the page from the live model tables.

#	Check	Result	Detail
1	Every DCS bubble of the simulator (BUB) is in the list	✓	34 / 34
2	Every instrument of the EPC P&ID register is in the list	✓	89 bubbles
3	Every ALM limit inside its instrument range	✓	19 / 19
4	Every trip setting inside the range of its sensor and = model K	✓	TSHH-100, TSHH-107, PSHH-101, LSHH-104
5	Fail position: makeCtrls() text = FAIL map = list = EPC P&ID	✓	11 / 11
6	Every C&E cause tag is in the list (tag or alarm id)	✓	45 / 45
7	Every control valve at design (model) within 10–90 %	✓	FV101 62 · FV103 50 · PV105 50 · FV104 45 · LV101 50 · TV102 50 · LV102 50 · LV104 50 · LV109 50 · PV110 50 · HV107 25
8	Sized valves within 10–90 % at D/H/L	✓	11 / 11
9	Orifices β 0.10–0.75 and Re ≥ 5 000	✓	FE-101 β 0.634 · FE-103 β 0.588 · FE-109 β 0.697
10	I/O totals add up (Σ rows = DCS + SIS = total)	✓	168 = DCS 110 + SIS 58
11	Flow ranges cover the 120 % case at ≤ 90 % of span	✓	FIC-101 70 % · FIC-103 74 % · FI-106 63 % · FI-110 75 % · FI-111 45 % · FI-109 46 % · FI-102 42 % · FI-105 37 %
12	Bed temperature range covers a cold bed	✗	range 100–250 °C (EQ-ei-5)
13	TV-102 passes the jacket cooling water at ≤ 90 %	✗	needs 114 % (EQ-ei-1)

9 · Engineering queries

ID	Query	Documents	Proposed resolution
EQ-ei-1	**TV-102 cannot serve both MP steam and cooling water** Sized on MP steam, TV-102 needs Cv 27.9 (rated 69.3); in COOL the same valve must pass the model's cooling-water flow K.Fcwj = 40 kg/s, which needs Cv 118 at 2 bar (assumed) — 1.7 × the steam valve. The model lets one valve pass either fluid at its own capacity (steam ≤ K.msj · x, water K.Fcwj · x).	model.js (TV-102 jacket), EPC P&ID sheet 2 (TV-102 + HY-102A/B), CHX-V1-IN-INL-001	Two valves on split range from TIC-102: TV-102A steam (Cv 69.3) and TV-102B cooling water (Cv ≈ 120); or keep HY-102A/B as route valves and give each route its own control valve. Model: no change needed for training; note it on the P&ID.
EQ-ei-2	**Model valve capacities differ from the sized valves** Equivalent Cv of the model constants at 100 % (same IEC equations) against the rated Cv selected here: FV-101 33 vs 69.3 (× 0.48); FV-103 177 vs 400 (× 0.44); HV-107 7.1 vs 3 (× 2.36). The model's valves are linear and sized to sit near 50 % at design; a real valve is chosen for the 120 % case at ≤ 85 % open with its own characteristic.	model.js K.Cv101…Cv110, FBmax, Fpa, msj; CHX-V1-IN-INL-001 part 4	Keep the model constants for training (they reproduce the HMB); the valve datasheets use the rated Cv of this list. Instructors read the DCS output as "% of the model capacity", not as stem travel.
EQ-ei-3	**Valve opening on the DCS is not the real stem position** The model's valve output is linear in flow; an equal-percentage valve of the rated Cv sits elsewhere at design: PV-105 model 50 % vs 77 %; TV-102 model 50 % vs 76 %; LV-109 model 50 % vs 67 %; PV-110 model 50 % vs 71 %; HV-107 model 25 % vs 59 %.	model.js step() (P.v), CHX-V1-IN-INL-001 part 4, SOP / Operating windows texts quoting openings	Quote valve openings in procedures as "% of capacity (model)"; on a real plant use the travel from the valve datasheet. No model change.
EQ-ei-4	**Instrument-air failure moves only the 11 control valves in the model** On loss of air the model drives FV/PV/LV/TV/HV-107 to the FAIL map; XV-101, XV-105, XV-104 (rec) close and EDV-101 opens on the P&ID (fail positions FC / FO assumed), and the HY route valves go to their FO/FC positions — the model keeps them where they were.	model.js step() FAIL map and air failure, EPC P&ID valve register, C&E CHX-V1-IN-CE-001	Confirm EDV-101 fail position (FO opens the loop to flare on any air loss: usually FO with a lock-in volume tank, or FC with SIS energise-to-open); add XV and HY valves to the model's air-failure response (lead, model.js).
EQ-ei-5	**R-103 bed temperature range starts at 100 °C** TI-107A…E (and TSHH-107) read 100–250 °C (IND, ui1.js). A cold or start-up bed (≈ 30 °C) is below the range: the transmitters go under-range and the hottest-of-five logic sees a "bad PV".	ui1.js IND ranges, CHX-V1-IN-INL-001, SRS (TSHH-107)	Calibrate TT-107A…E 0–300 °C (normal 158.5 °C at 53 % of span; trip 220 °C inside); change IND r in ui1.js (lead).
EQ-ei-6	**Line sizes at the instruments differ between the EPC P&ID and the line list** FV-101 / FT-101: P&ID 4"-P-1002 vs line list 3"-P-1002-B1; FV-104 / FT-104 (E-101 leg): P&ID 12"-SL-1015 vs line list 16"-P-1005-B3; LV-101: P&ID 6"-SL-1014 vs line list 4"-P-1008-B3; PV-105 / FT-106: P&ID 8"-G-2009 vs line list 6"-H-2010-B1; HV-107 / FT-107: P&ID 2"-G-2010 vs line list 1-1/2"-H-2011-B2; LV-104 / FT-110: P&ID 6"-P-2011 vs line list 4"-P-1013-B1; TV-102 (steam): P&ID 4"-MS-2020 vs line list 6"-S-3003-A1. The P&ID used velocity limits of its own before CHX-V1-PE-LL-001 existed; this list and the valve sizing use the line list.	EPC P&ID CHX-V1-PID-001 (line labels), CHX-V1-PE-LL-001	Re-issue the P&ID line labels from CHX-V1-PE-LL-001 (numbers and sizes) at rev 1.
EQ-ei-7	**HV-113 permissive has no signal** The benzene start permissive "HV-113 open" (C&E, purgePermissives()) reads the position of a hand valve. A hand valve gives the SIS nothing to read.	C&E CHX-V1-IN-CE-001, model purgePermissives(), EPC P&ID sheet 1	Add limit switch ZSO-113 (SIS DI) on HV-113, or replace the permissive by a CW flow/temperature measurement on E-103 (HAZOP R06 style).
EQ-ei-8	**LV-104 letdown cavitates and releases dissolved hydrogen** LV-104 drops the product from 30.0 to 3.6 bar a (ΔP 26.4 bar). With a standard globe trim (FL 0.9) the choked limit is 24.0 bar, below the ΔP: the valve would choke and cavitate; the multi-stage anti-cavitation trim chosen here (FL 0.95, typical) raises the limit to 26.8 bar, and the cavitation index xF 0.89 is still above its incipient value. The liquid is also saturated with hydrogen at 29 bar g, so gas comes out of solution in the valve and the line to the tanks. The model treats LV-104 as a plain √ΔP valve.	model.js (L4), CHX-V1-PE-LL-001 4"-P-1014-A2, CHX-V1-IN-INL-001	Multi-stage anti-cavitation trim, hardened; outlet body and line sized for the released gas; keep the tank vent/flare route (EQ-ep-12).
EQ-ei-9	**FV-101 pressure drop rests on a flat pump curve** The model has no P-100 curve (feed = K.FBmax · x unless PI-101 ≥ 36 bar a), so the FV-101 ΔP here is the eproc estimate "shut-off − required" (4.53 bar at design). A real centrifugal pump loses head with flow and the valve ΔP at 120 % is smaller.	model.js (FB), CHX-V1-PE-LL-001 part 6, P-100 datasheet	Fix the P-100 rated point and curve on its datasheet and re-size FV-101 (Cv margin is large: 75 % at 120 %).

10 · Assumptions

Item	Value
Instrument signals	4–20 mA HART for analogue; 24 V DC dry contacts for status; solenoids 24 V DC, de-energise to trip (assumed, typical)
Systems	DCS for control, indication and BPCS on/off valves; SIS-100 (safety PLC) for trips, permissives and SIS valves; DCS ↔ SIS serial link for repeats (typical)
Spare I/O	20 % installed spare per system and signal type, rounded up (typical practice)
Ranges	Existing tags: the model's ranges (makeCtrls() lo/hi, IND r in ui1.js). New tags: the range of the DCS display they feed; recommended tags by the rule normal at 50–70 % of span
Instrument types	Typical choices for the service (fluid, temperature, slurry, SIS diversity); the model does not fix them except the R-101 elements, which the model names thermocouples
Valve sizing	IEC 60534-2-1 (ISA-75.01) liquid and gas equations; Fp = 1 (body = line or reducer losses neglected); FL, xT, xFz typical for the valve style; equal % rangeability 50; rated Cv from typical body Cvs and reduced trims (none for slurry); governing case ≤ 85 % open; every case 10–90 %
Valve ΔP	From CHX-V1-PE-LL-001 part 6 (hydraulic budgets) and the model's pressures at each HMB case; line losses of the line list removed; TV-102 jacket pressure = saturation at T_2 + Q/UAj (model); R-102 jacket CW case 2 bar across the valve (assumed)
Critical pressures	benzene 48.9, cyclohexane 40.7, water 220.6 bar a (typical data); vapour pressures from the model (PsatC, PsatW)
Flow element	ISO 5167-2 orifice, flange tappings, Reader-Harris/Gallagher C; ΔP at the top of the transmitter range 250 mbar (typical), 1 000 mbar for FE-101 to keep β ≤ 0.75; gas expansibility ε (ISO 5167-2)
Steam and gas	k = 1.30 (steam), 1.40 (hydrogen-rich gas); steam Z = 0.95 (eproc); ideal gas density at the valve inlet
Valve styles (typical)	globe, contoured plug: FL 0.9, xT 0.7, xFz 0.5 · eccentric rotary plug (slurry): FL 0.85, xT 0.61, xFz 0.45 · three-way globe, two ports: FL 0.85, xT 0.7, xFz 0.45 · globe, multi-stage anti-cavitation trim: FL 0.95, xT 0.8, xFz 0.75 · globe, cage-guided low-noise trim: FL 0.9, xT 0.72, xFz 0.5
Body Cv table (typical)	1" 12 · 1-1/2" 28 · 2" 46 · 3" 110 · 4" 190 · 6" 400 · 8" 700 · 10" 1100 · 12" 1600 · 14" 2100 · 16" 2800 · 18" 3600 · trims 1, 0.63, 0.4, 0.25
TV-102 cooling-water case	supply 5.01 bar a, 2 bar across the valve (assumed)